# *Intelligent Carbon Accounting Chatbot Using Retrieval-Augmented Natural Language Processing*

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## *Abstract*

Climate regulations mandate increasingly complex carbon accounting and disclosure requirements. Organizations face rapid changes in standards like the UK Streamlined Energy and Carbon Reporting (SECR), EU Emissions Trading System (ETS), and the Greenhouse Gas (GHG) Protocol. Professionals struggle to navigate lengthy, evolving regulatory documents to answer compliance queries efficiently. This project proposes an intelligent question-answering chatbot using retrieval-augmented generation (RAG) techniques and transformer-based natural language processing (NLP) models. The system will extract accurate, cited answers from authoritative carbon accounting texts, reducing manual search time and compliance risk. Key objectives include corpus curation, fine-tuning domain-specific QA models, developing an interactive web interface, and achieving ≥85% answer accuracy with full source citation. The artifact will demonstrate practical utility for sustainability teams, auditors, and consultants navigating the fast-changing carbon reporting landscape.

# 1. Introduction

## 1.1 Background on Carbon Accounting and Regulatory Complexity

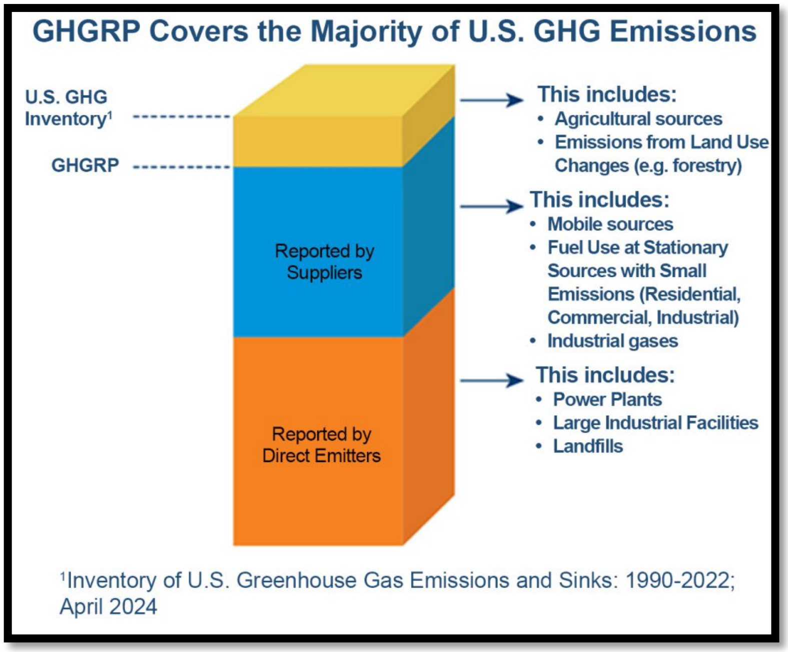


**Figure 1 : Carbon Emissions Reporting**

Carbon accounting has emerged as a critical business function driven by climate policy, investor expectations, and corporate sustainability commitments. Regulations such as the UK SECR (introduced in 2019), the EU ETS (reformed in 2023), and voluntary frameworks like the GHG Protocol require detailed quantification, reporting, and verification of greenhouse gas emissions. These standards evolve frequently—updating emission factors, reporting scopes, and calculation methodologies—creating a dynamic compliance environment. Organizations must track multiple overlapping regulations across jurisdictions, each with specific reporting timelines, data requirements, and penalties for non-compliance.

## 1.2 The Problem: Information Overload and Manual Search

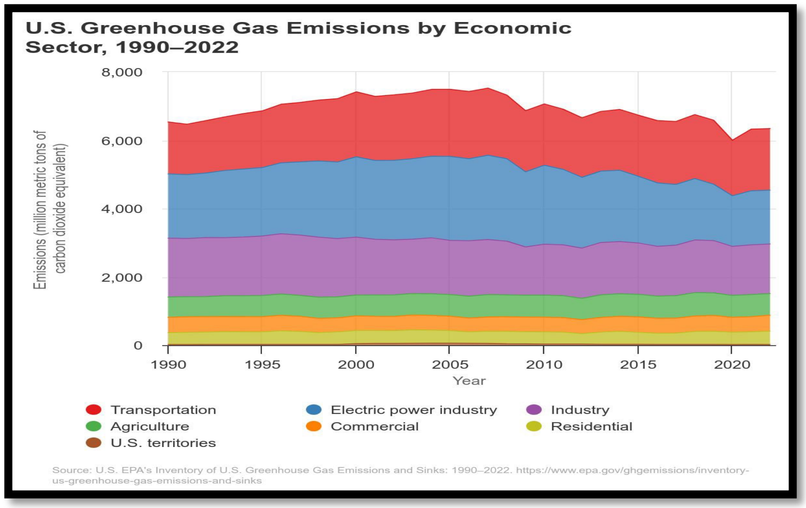


**Figure 2 : Growth in GHG activity.**

Compliance professionals currently rely on manual searches through lengthy PDF guidance documents, government portals, and industry commentaries to answer routine questions such as: "What are the latest Scope 2 emission factors for electricity?" or "How should we report biogenic CO2 under SECR?" This process is time-consuming, error-prone, and inefficient—especially when regulations change mid-reporting cycle. Misinterpretation of standards can lead to reporting errors, regulatory penalties, and reputational damage. There is a clear need for intelligent, automated tools that deliver accurate, up-to-date answers with verifiable source citations.

## 1.3 Project Objectives

This project aims to develop a domain-specific NLP chatbot for carbon accounting queries using retrieval-augmented generation. The system will:

- Curate and preprocess a corpus of authoritative carbon accounting texts (regulatory standards, guidance documents, FAQs).

- Implement a two-stage retrieval-augmented QA pipeline combining document search and transformer-based answer generation.
- Build an interactive web-based chatbot interface for user queries.
- Evaluate system performance using accuracy metrics (Exact Match, F1 Score), citation quality, response latency, and user satisfaction.
- Address ethical, legal, and environmental considerations related to AI-driven regulatory advice.

## *2. Literature Review*

### *2.1 Evolution of Question Answering Systems in NLP*

Question answering (QA) has progressed from rule-based systems and keyword matching to sophisticated neural architectures. Traditional IR-based QA relied on document retrieval followed by extractive heuristics.

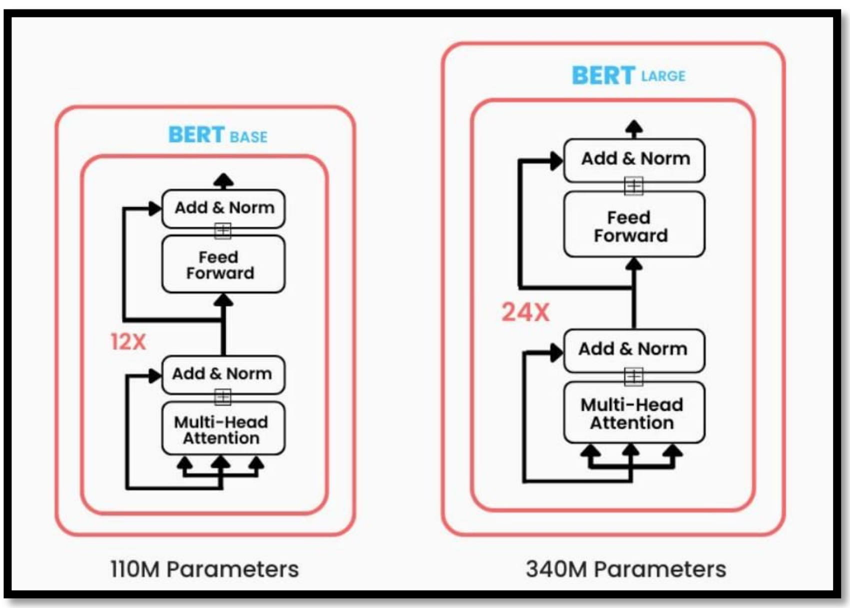


**Figure 3 : BERT Architecture**

The introduction of transformer models—particularly BERT (Devlin et al., 2018) and its variants—revolutionized QA by enabling contextual understanding of queries and passages. BERT's bidirectional attention mechanism captures semantic relationships, improving answer extraction accuracy.

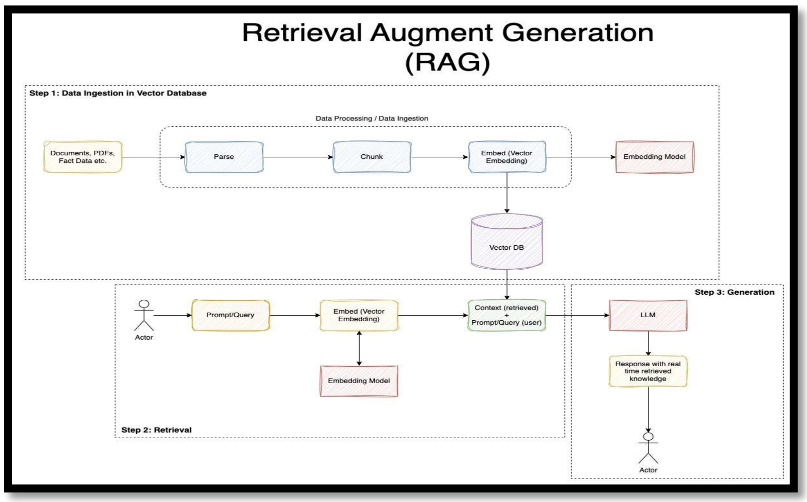


**Figure 4 : Standard RAG Pipeline**

More recent innovations like Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) combine dense passage retrieval with generative models (e.g., BART, T5) to synthesize answers from multiple sources, addressing limitations of purely extractive or generative approaches.

## *2.2 Applications of NLP in Legal and Regulatory Domains*

NLP chatbots have been deployed in legal, healthcare, and financial compliance domains with promising results. Systems like CaseText and ROSS Intelligence use transformer models to query legal databases and extract relevant case law. In healthcare, clinical decision support tools leverage BioBERT and Med-PaLM for evidence-based recommendations (Singhal et al., 2023). However, carbon accounting—despite its regulatory importance—remains under-explored in NLP research. Existing sustainability software focuses on data collection and visualization but lacks intelligent query interfaces for regulatory interpretation.

### *2.3 Challenges: Hallucination, Domain Adaptation, and Citation*

A critical challenge for generative models is hallucination—producing plausible but factually incorrect answers. In regulatory contexts, even minor inaccuracies can lead to compliance failures. Retrieval-augmented approaches mitigate this risk by grounding answers in retrieved documents. Domain adaptation is also essential: general-purpose language models lack specialized vocabulary and regulatory context. Fine-tuning on domain-specific corpora improves performance (Lee et al., 2020). Finally, explainability and citation are paramount for regulatory applications—users must verify answer sources for auditability and trust.

### *2.4 Gaps in Current Research*

No existing system addresses the specific needs of carbon accounting compliance: real-time updates to regulatory texts, multi-jurisdictional coverage, and cited answers for audit trails. This project fills that gap by building a specialized retrieval-augmented chatbot tailored to the carbon accounting domain.

# 3. Proposed Methodology

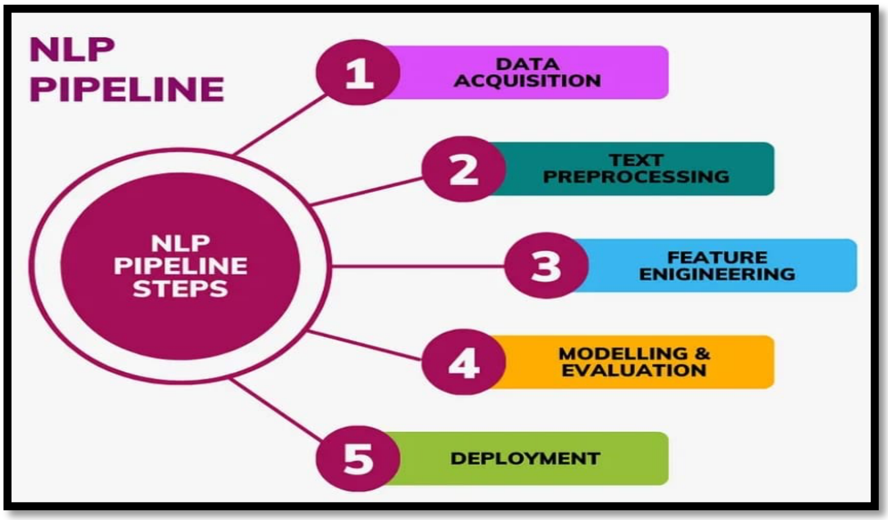


**Figure 5 : NLP Pipeline Steps**

## 3.1 Data Collection and Preprocessing

Data Sources: The corpus will comprise approximately 500-1000 authoritative documents including:

- UK Government SECR guidance and consultation documents
- EU ETS Directive texts and implementing regulations
- GHG Protocol Corporate and Scope 2 Guidance
- ISO 14064-1:2018 standard
- National emission factor databases (DEFRA, EPA)
- Industry FAQs and technical notes from professional bodies (IEMA, CDP)

Data Acquisition: Documents will be downloaded from official government portals, standards bodies, and open-access repositories. Web scraping will comply with terms of service and robots.txt protocols.

Preprocessing Pipeline:

- PDF-to-text conversion using OCR tools (Tesseract, pdfplumber) with quality checks for extraction accuracy.
- Text cleaning: Remove headers, footers, page numbers, and boilerplate content.
- Segmentation: Divide documents into semantically coherent passages (150-300 words) suitable for retrieval.
- Annotation: Tag passages with metadata (document source, regulation type, publication date, section headings).
- Terminology normalization: Map synonymous terms (e.g., "GHG emissions" ↔ "carbon emissions") using domain dictionaries.

## *3.2 Model Selection and Implementation*

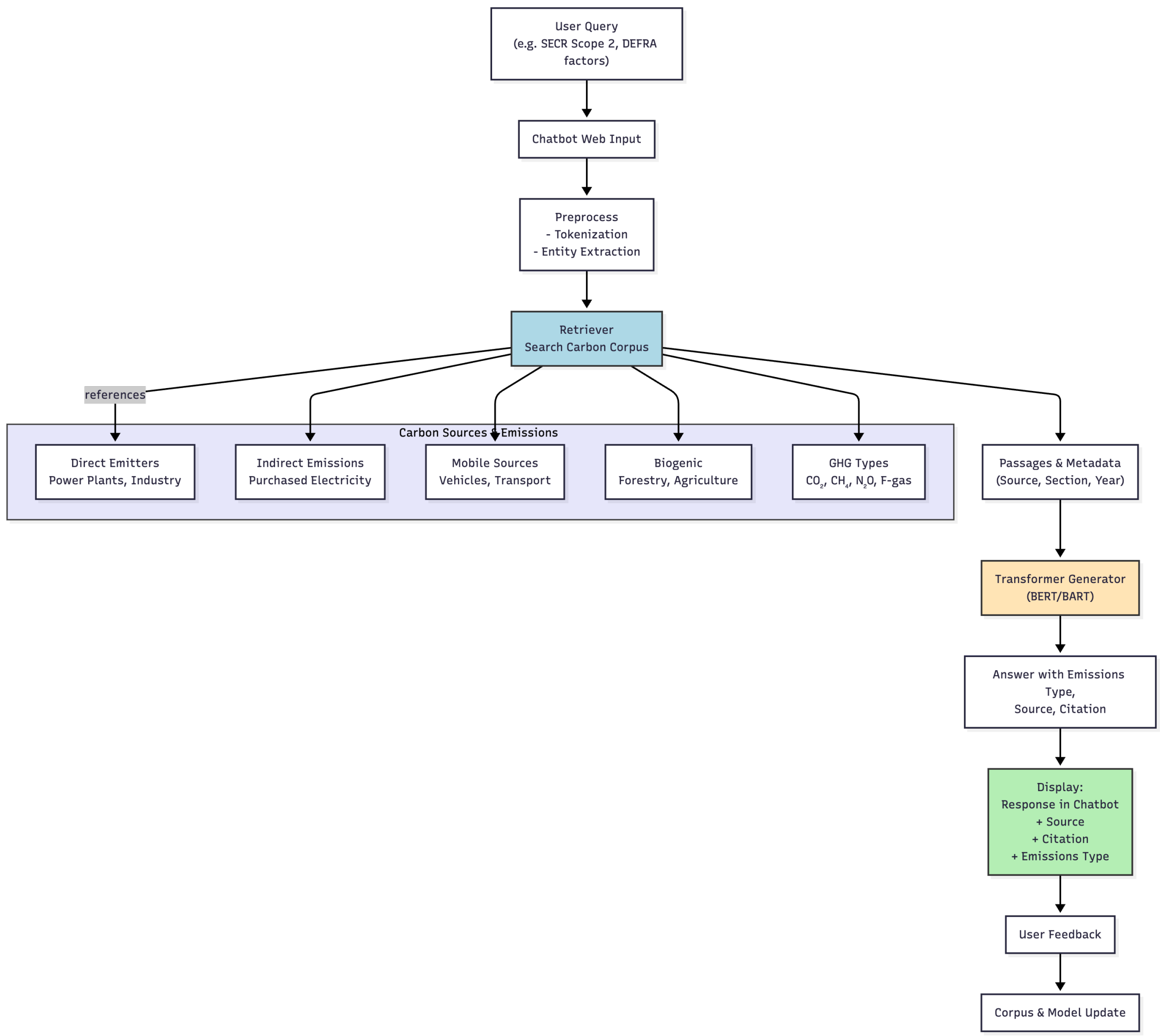


**Figure 6 : Our Problem Specific Pipeline**

***Retrieval Module***: Dense Passage Retrieval (DPR) or BM25+BERT hybrid will index preprocessed passages. DPR uses dual-encoder architecture to embed queries and passages in shared semantic space, enabling efficient similarity search via FAISS or Elasticsearch.

***Answer Generation Module***: Fine-tune a transformer-based model (DistilBERT for efficiency, or BART/T5 for generative capability) on carbon accounting QA pairs. Training data will include synthetic question-answer pairs extracted from regulatory texts plus manually annotated examples for quality assurance.

***Citation Mechanism***: The system will output answers alongside source document references (regulation name, section, page number) to enable verification and audit compliance.

***Implementation Tools***:

- Programming Language: Python 3.9+
- NLP Libraries: HuggingFace Transformers, spaCy, NLTK
- Retrieval: Elasticsearch or FAISS for vector search
- Web Framework: Flask or Streamlit for chatbot UI
- Deployment: Docker containerization, cloud hosting (AWS/Azure free tier)

### *3.3 Evaluation Metrics*

***Quantitative Metrics***:

- Exact Match (EM): Percentage of answers exactly matching ground truth.
- F1 Score: Token-level overlap between predicted and reference answers.
- Citation Accuracy: Percentage of answers with correct source attribution.
- Response Latency: Average time from query submission to answer display (target <2 seconds).

***Qualitative Evaluation***:

- Expert Review: Carbon accounting professionals assess answer correctness, relevance, and clarity.
- User Satisfaction Survey: Likert-scale feedback on usability, trust, and perceived value.
- Usability Testing: Task completion rates and time-on-task for common queries.

# *4. Objectives*

## *4.1 Project Goals*

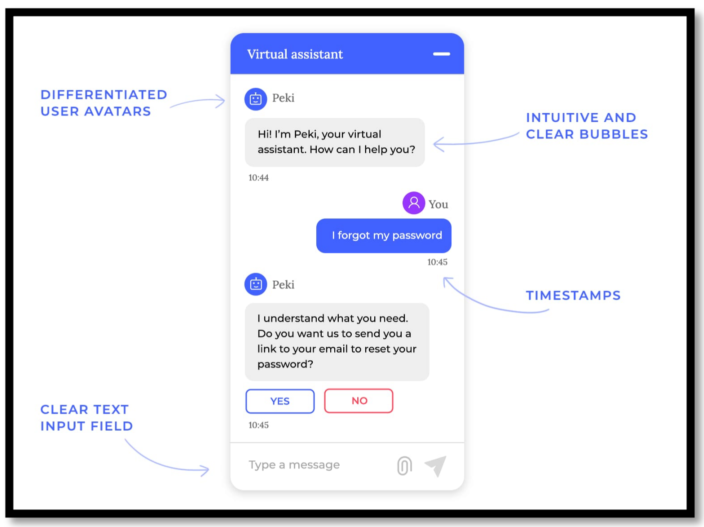


**Figure 7 : Chat User Interface**

The project will deliver a working prototype demonstrating the feasibility of retrieval-augmented NLP for regulatory compliance in carbon accounting. Success criteria include:

- ***Corpus Creation***: Assemble and preprocess 500+ authoritative carbon accounting documents.

- ***Model Training***: Fine-tune retrieval and generation models achieving ≥85% F1 score on test queries.
- ***System Integration***: Build end-to-end QA pipeline with citation output.
- ***User Interface***: Deploy web-based chatbot with intuitive query interface.
- ***Evaluation***: Validate system performance through quantitative metrics and expert feedback.

### *4.2 Measurable Outcomes*

- Answer Accuracy: ≥85% F1 score, ≥70% Exact Match on held-out test set.
- Citation Rate: 100% of answers include source document reference.
- Response Speed: Average latency <2 seconds per query.
- User Satisfaction: ≥80% positive feedback from pilot testers.

## *5. Proposed Artifact*

The final deliverable is an interactive web-based chatbot system comprising:

- Backend QA Engine: Retrieval-augmented pipeline (document indexing, query processing, answer generation, citation extraction).
- Frontend Interface: Clean, responsive web UI built with Streamlit or Flask, allowing users to input queries and view cited answers.
- Dashboard Features: Display recent regulatory updates, frequently asked questions, and answer confidence scores.
- Documentation: Technical documentation covering system architecture, API endpoints, deployment instructions, and maintenance procedures.

- Demo Video: Walkthrough demonstrating system capabilities on representative carbon accounting queries.

The system will be designed for extensibility—future versions could incorporate multi-domain coverage (tax, company secretarial) or real-time regulatory monitoring via automated document ingestion pipelines.

# 6. Ethical, Legal, and Environmental Considerations

## 6.1 Accuracy and Liability

The system will include disclaimers clarifying it provides informational guidance, not legal advice. Users are responsible for verifying answers against official sources. Citations enable auditability and reduce risk of misapplication.

## 6.2 Data Privacy and Security

The corpus consists solely of publicly available regulatory texts—no personal or proprietary data is processed. User queries will be logged anonymously for system improvement, with opt-out available.

## 6.3 Environmental Impact

Model training and deployment will prioritize energy efficiency: use of smaller models (DistilBERT), serverless deployment, and caching frequently requested answers to reduce computational overhead.

## 6.4 Bias and Fairness

The system reflects biases in source documents (e.g., jurisdiction-specific regulations). Documentation will clearly state coverage scope and limitations to prevent misuse in contexts outside the training data distribution.

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